

Optimization

0. Syllabus and course organization

José Firmino Aguilar Madeira

R. Prerequisites and review

Support material, not a chapter in the course progression: used mainly from Ch. 3 onwards.

- Geometry: vectors, angle, line, segment, hyperplane
- Differential calculus: level sets, directional derivative, gradient, Hessian
- Linear algebra: eigenvalues and eigenvectors, positive definite matrices, Sylvester's criterion
- Convexity of sets and of functions

1. Introduction to optimization

1.1 Optimization problems and modelling

1.2 Applications of optimization: where it is used — and what they have in common

2. Univariate optimization

2.1 Univariate optimization: definition, classification and general remarks

2.2 Golden section search

2.3 Successive quadratic interpolation (Powell's method)

2.4 Bisection method

2.5 Newton–Raphson method

2.6 Numerical approximation of derivatives

3. Unconstrained multivariate optimization

3.1 Multivariate optimization: definition and general remarks

3.2 Derivative-free methods

3.2.1 Random search

3.2.2 Grid search

3.2.3 Nelder–Mead simplex method

3.2.4 Box's method — a simplified version of EVOP (evolutionary operation)

3.2.5 Hooke–Jeeves method (pattern search)

3.3 Derivative-based methods

3.3.1 Steepest descent method

3.3.2 Newton's method

3.3.3 Conjugate gradient method (Fletcher–Reeves)

4. Constrained optimization

4.1 Constrained extrema: Lagrange multipliers

4.2 Karush–Kuhn–Tucker (KKT) conditions

4.3 Penalty methods

4.3.1 Exterior penalty function method (EPFM)

4.3.2 Barrier / interior penalty function method (IPFM)

4.3.3 Constraints in practice: penalize, reject or compare

5. Global optimization

5.1 GLODS — Global and Local Optimization using Direct Search

5.2 Simulated annealing

5.3 Genetic algorithms

6. Multiobjective optimization

6.1 The multiobjective problem

6.2 Pareto front

6.3 Aggregation (scalarization) methods

6.4 The ε -constraint method

6.5 Direct MultiSearch (DMS)

6.6 NSGA-II (multiobjective genetic algorithms)

Main

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009.
- Miettinen, K., *Nonlinear Multiobjective Optimization*, Kluwer Academic Publishers, 1999.

Supplementary

- Arora, J. S., *Introduction to Optimum Design*, 3rd ed., Academic Press, 2011.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012.
- Deb, K., *Multi-Objective Optimization Using Evolutionary Algorithms*, Wiley, 2001.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987.
- Horst, R., Tuy, H., *Global Optimization: Deterministic Approaches*, 3rd ed., Springer, 1996.
- Jahn, J., *Vector Optimization: Theory, Applications, and Extensions*, 2nd ed., Springer, 2011.

Lecturer's theses (supplementary material; the first in Portuguese)

- Madeira, J. F. A., *Optimização multi-objectivo da topologia de estruturas com algoritmos genéticos*, PhD thesis in Mechanical Engineering, IST, Universidade Técnica de Lisboa, 2004.
- Madeira, J. F. A., *Multiobjective and global optimization with directional direct search*, PhD thesis in Mathematics, IST, Universidade de Lisboa, 2019.

Papers on the course methods

- Custódio, A. L., Madeira, J. F. A., *GLODS: Global and Local Optimization using Direct Search*, Journal of Global Optimization 62(1):1–28, 2015.
- Custódio, A. L., Madeira, J. F. A., Vaz, A. I. F., Vicente, L. N., *Direct multisearch for multiobjective optimization*, SIAM Journal on Optimization 21(3):1109–1140, 2011.
- Custódio, A. L., Madeira, J. F. A., *MultiGLODS: global and local multiobjective optimization using direct search*, Journal of Global Optimization 72(2):323–345, 2018.

Code: the course MATLAB and Python code is in the course repository; GLODS and DMS are distributed in class. Each deck ends with “References and resources”; full list: *Course bibliography*.

Assessment has a theoretical component and a practical component.

- **Theoretical component (NT):** 2 tests during the teaching period (minimum mark 8 out of 20 in each, minimum average 9.5) or an exam (minimum mark 9.5).
- **Practical component (NP):** a project with compulsory presentation and discussion (minimum mark 9.5).

Final mark

$$NF = 0.7 NT + 0.3 NP$$